# JS Astrophysics Computational Lab: Assignment 3 Part 3

Prof. Johanna Vos

February 17, 2026

## Lab Objectives

This document contains the last part of the third assignment for the module. In Parts 1 and 2, you analysed the Gaia Catalog of Nearby Stars (GCNS) to find close stellar flybys in the past and future. In this lab, we will use another data source, the Ultracool Sheet, to search for flybys that are missed by Gaia. As outlined in last week's lab, your final report will include your analysis from Parts 1, 2 and 3 of this assignment.

## Sources Hidden from Gaia

While the Gaia Mission provides measurements for a very large sample of sources in our Milky Way, it misses two particularly important populations of objects: the lowest-mass stars and the brown dwarfs. This is because, at lower temperatures ($< 2500$ K), sources start to emit at longer wavelengths. Figure 1 shows a range of blackbody spectra for a range of different temperatures. The Gaia Mission operates at optical wavelengths, from $\sim 330 - 1000$ nm. Stars with effective temperatures above $\sim 3000$K emit a large amount of flux at these wavelengths, whereas stars and brown dwarfs with colder temperatures emit little or no flux at these wavelengths.

### M Dwarfs

With temperatures of $\sim 2500 - 3500$K, M dwarfs are the coolest stellar spectral type. Many M dwarfs have measured photometry in the Gaia sample, but in many cases they do not have a measured radial velocity. This is because a spectrum is required for a radial velocity measurement, which requires even more photons than a photometric point. For example, the exoplanet host Trappist-1 is included in the GCNS, but it does not have a radial velocity, and thus we could not determine its closest approach in the last lab.

Over the past decade or so, M dwarfs have been targeted by many exoplanet searches using the transit and radial velocity techniques (e.g. Trifonov et al. 2018, Ghachoui et al. 2023). M dwarfs are particularly suitable targets to detect temperate, low-mass rocky planets due to their relatively small mass and radii, which facilitates the detection of small planets, and because their habitable zones are located closer-in than for hotter stars. For example, of Trappist-1's seven planets, three of them are located inside the habitable zone. Since they are missed by Gaia, it is important to study them with other instruments.

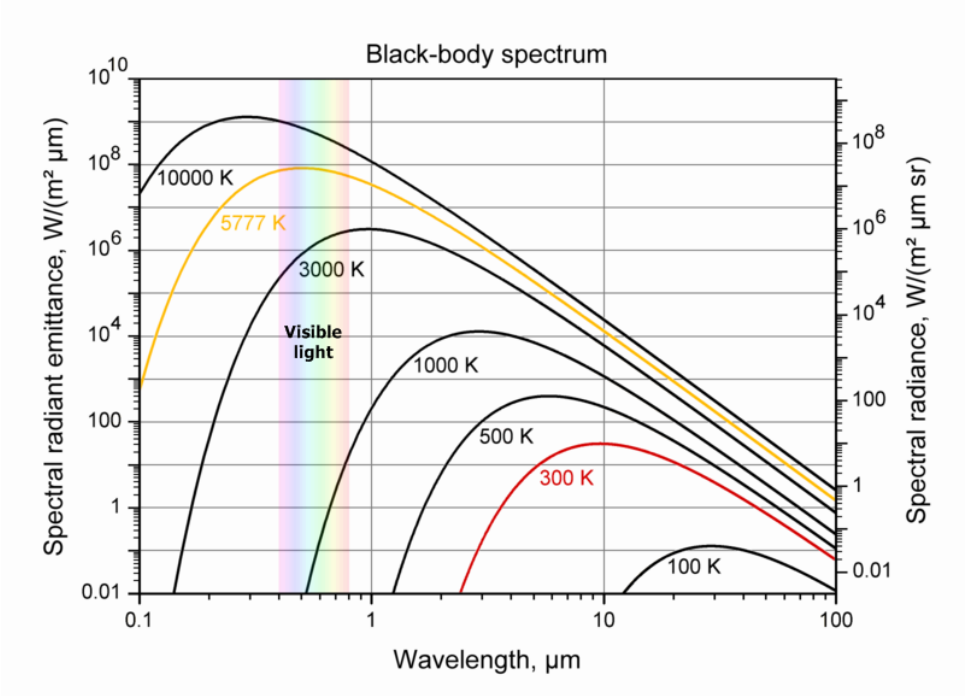



Figure 1: Blackbody spectra for a range of temperatures. Hotter stars emit singificant flux at optical wavelengths, so Gaia can easily observe them. M dwarf stars and brown dwarfs emit little to no flux in the optical and so must be studied in the infrared.

## Brown Dwarfs

Brown dwarfs, with temperatures below 2500K, are generally missed by Gaia, since they are simply too faint in the optical bands to be detected. Brown dwarfs are substellar objects whose masses fill in the gap between the most massive exoplanet and the least massive stars ($\sim 13-80\ M_{\rm Jup}$). Brown dwarfs are thought to form in a similar manner to stars: via the collapse of a giant molecular cloud of gas and dust. However brown dwarfs do not gain sufficient mass to sustain hydrogen burning in their cores. While stars burn hydrogen for millions (or even billions) of years, brown dwarfs do not produce energy, and thus spend their lives cooling and dimming. This means that brown dwarfs reach temperatures much colder than stars, and are more similar to planets. For example, the coldest brown dwarf that has been discovered to date has an effective temperature of $\sim 250$ K – similar to the North Pole on a warm day.

The Hertzsprung-Russel Diagram breaks down once we reach substellar masses. At these lower temperatures, colour is no longer a proxy for the temperature, as is the case for higher mass stars. Instead, we tend to plot a "colour-magnitude" diagram for substellar objects, shown in Figure 2, which shows how their colours change as brown dwarfs cool and evolve through spectral types M, L, T and Y.

When brown dwarfs were discovered, new spectral types L, T and Y were introduced to categorise them according to temperature. At temperatures $< 2200$K, brown dwarfs enter the L spectral type. At these temperatures, molecules such as water and carbon monoxide form abundantly in their atmospheres. Additionally, clouds composed of molten iron droplets and magnesium silicates start to form. In contrast, T dwarfs appear relatively cloud free, and methane begins to form. Finally, Y dwarfs are the coldest spectral type. These atmospheres are so cold that clouds made of water and ammonia start to from, like

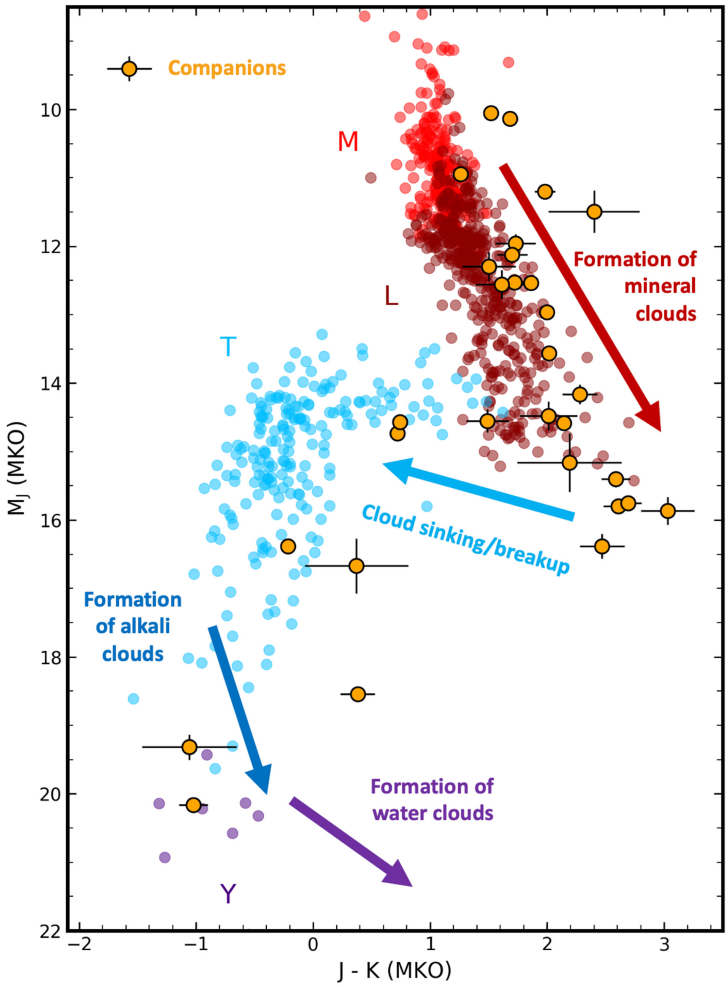



Figure 2: Colour-magnitude diagram showing the brown dwarfs and imaged companions from Gao et al. 2021.

those we see on Earth and Jupiter respectively. Since their temperatures overlap with those of exoplanets but can be observed much more easily, brown dwarfs are often use as analogs for exoplanets. Studying their atmospheres provides crucial context for our understanding of exoplanetary atmospheres.

## The Ultracool Sheet

Late-type stars and brown dwarfs are more easily observed and studied at infrared wavelengths. In order to measure radial velocities for these objects, we must use ground-based near-IR spectrographs, and take each spectrum one-by-one. Thankfully, a group of scientists maintain a catalog called The UltraCool Sheet (Best et al. 2020), in which they have collated data for 3,000+ so-called "ultracool dwarfs" (spectral types M6 and later) and directly imaged exoplanets. Like Gaia, the catalog contains a mix of photometry, astrometry, kinematic information and spectral information. Although the sample is much smaller, we can use this dataset to search for stellar flybys from less massive sources.

Our aim is to identify low-mass flybys using the Ultracool Sheet to add to our Gaia DR3 plot. Approximating masses for these objects is significantly more difficult due to a degeneracy between age and mass that emerges in the substellar regime, so we will not estimate impulses for this sample.

# Part 3 Tasks:

1. Download and read in the Ultracool Sheet. The up-to-date Ultracool Sheet is available here as a google spreadsheet. The `README` tab contains explanations of each of the columns. How many objects are in the sample?

2. Make a plot of the absolute $J$-band against the $J - K$ colour, and highlight the different spectral types as well as the small sample of imaged exoplanets that are in the sample. You should get a plot similar to Figure 2. You should get a similar plot whether you decide to use the 2MASS or MKO $J$ and $K$ filters. The spectral type is contained in the `sptnum_formula` column.

3. We are going to use the linear approximation equations again to find less massive flybys. Save the relevant columns as `numpy` arrays (or another table and/or array type). The best astrometric and kinematic values and their associated errors are stored in the columns with "`_formula`" in the name: e.g. `rv_formula`, `pmdec_formula`, `pmra_formula`, `plx_formula` etc. These give the best available measurement from the literature, as chosen by the Ultracool Sheet team. You should also store values such as the source name and spectral type for later analysis of your flybys.

4. How many sources in this sample have measured RVs?

5. Calculate the perihelion distance and time for the sample, and plot your results on a $d_{ph}$ versus $t_{ph}$ plot (zoomed in on only the closest approaches). As before, highlight suitable distances of interest on your plot.

6. What is the closest approaching ultracool dwarf? Discuss its trajectory in comparison to the closest approaching sources from your Gaia sample. Overplot this target on the color-magnitude diagram to roughly approximate its spectral type.